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Vestel:
Improving Distribution Management and 3PL Relations



*This case was written by **Professor Baris Tan** and **Assistant Professor Zeynep Aksin**, with the assistance of Hülya Gökbel, MBA 2004, Koç University, Istanbul. It is intended to be used as the basis for class discussion rather than to illustrate either effective or ineffective handling of a management situation.*

The case was made possible by the co-operation of Vestel Durable Consumer Goods.

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It was a sunny spring morning, and Levent Hatay, the General Manager of Vestel Durable Goods Marketing, was driving to work. Given the construction on his regular route, he was forced to drive through Sarıyer. On the main shopping street, he passed the Vestel showroom closely followed by those of Bosch, Siemens, and Beko. All of them had little vans parked in front of them, making deliveries of TVs, DVDs, Refrigerators, etc. He noticed that the van in front of the Vestel store was half empty. He thought*: It is unbelievable, we all have stores on the same street, we sell similar kinds of products, we all have daily deliveries of these products, yet each of us has a separate van making the deliveries. Couldn't we reduce our transportation costs if we worked together? We use a third party logistics company anyway. Would the economics justify cooperating with our own competitors? I could use some more cost reduction in transportation and logistics, I wonder what Beko or Siemens would think?*

Company background

Zorlu Holding

The foundations of Zorlu Holding were laid in the early 1950s. It has since become one of the major conglomerates in the Turkish economy operating in various economic segments including textiles, financial services, electronics and energy with its 22 thousand employees and 46 companies. The total revenue of the group excluding DenizBank exceeded 2.1 billion in 2002 while export of the group increased from USD 600 million in 1998 to USD 1.43 Billion in 2002.

Vestel Group of Companies

In 1994, Zorlu Holding acquired all shares of Vestel Electronics. Vestel is the flagship company of the electronics, consumer durables and IT Group comprising of 16 companies, 7 domestic and 9 in Europe and overseas. It operates in three major business categories – electronics, household appliances and information technology.

Vestel Electronics A.S. is the largest electronics manufacturer in Turkey. Its core product TVs were accounting for 70% of total sales in 2000 and monitors represented 5%. In 2001, Vestel Electronics produced a total of 4.6 million televisions, making up 65% of the country's total TV production. In 2002, TV production increased to 6.4 million.

While being a leading brand in the Turkish television market with 30% market share as of year 2002 Vestel Electronics is also the largest domestic exporter with 65% share. Being the largest full-range television ODM (Original Design and Manufacturing) in Europe, Vestel Electronics had a market share of 17% in OEM sales. It is the major supplier to large retailers such as Carrefour, Makro, Metro, Crown, Quelle, Hitachi, JVC, and Toshiba. The television production plant in Manisa, Turkey, has the world's largest single plant television production capacity of 8 million units. Additional facilities are under construction outside of Moscow with yearly capacity of 1.5 million units.

Vestel Communication, the consumer electronics manufacturer of the Vestel Group of Companies, has the largest component surface-mounted assembly (SMD) capacity in the country. Using SMD machinery in Turkey, Vestel Communication currently produces DVD players, analogue and digital receivers, personal computers and Internet access devices of nearly 4 million pieces per year.

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Vestel White was established in late 1997 within the Vestel campus in Manisa to produce and sell household appliances, mainly refrigerators. Refrigerator production began in June 1999, followed by air conditioners in 2000 and washing machines in 2003. Capacities are world-scale with 2 million refrigerators, 500,000 air conditioners, and 1 million washers per year.

Vestel Foreign Trades, the second largest export company in 2002, carries the export of Vestel Group products. Vestel Foreign Trade carries out the marketing and sales of all products manufactured by Vestel Electronics, Vestelkom and Vestel White in international markets. In 2002, total exports reached USD 1.078 million, of which 80% was sold to EU countries, while the remaining 20% was exported to Eastern Europe, North and South Africa, the USA, Russia, the Balkan states, Australia, the Middle East and the Turkic Republics.

Vestel Durable Goods Marketing is the marketing, sales and distribution company of the Vestel Group in Turkey. It is engaged in the domestic sales of industrial products manufactured by Group subsidiaries and its own Vestel branded products purchased from world producers under the Vestel brand name. Exhibit 1 gives the total annual revenues of Vestel group companies.

The product portfolio of the Company includes electronic equipment, hardware, IT products, white goods, brown goods, air conditioners and small appliances. Established in 1986, the Company has one of the largest distribution networks in the consumer durables sector in Turkey with a total of 1500 sales outlets and 650 after-sales service centers. Of the sales outlets, 700 are exclusive stores and showrooms. Vestel Durable Goods Marketing ranks in the top echelons of all well-known brands in its sector. In 2003, Vestel started distributing a number of different international brands including JVC cameras and audio, Moulinex branded small appliances, Zanussi branded built-in appliances. In 2003, revenues are expected to exceed USD 320 million.

The Durable Goods Market in Turkey

The Turkish durable goods market includes major producers such as Arçelik, Beko, Vestel, Profilo, Siemens, Bosch, Ariston, and Indesit. Turkish television producers are the leading suppliers of Europe and they satisfy 35% of the demand for televisions in Europe with an export of 12 million units per year in 2002. Turkish white goods producers are ranked sixth in production and export mainly to Europe.

Domestic competition in this market is intense, populated by both local and global players. Well known competitors for Vestel are Beko, Profilo, Siemens, Bosch, and Philips (Exhibit 7). The durable goods business in Turkey is different than in other countries making it difficult for globally successful companies to compete and succeed in the Turkish market. Only after buying the local company Profilo could Bosch/Siemens enjoy any success in Turkey. Sony and Philips' market share are still around five percent. Why is this so? According to Levent Hatay, there are some barriers to entry which can be summarized as follows:

Production: Production of the major appliances such as TVs, refrigerators, washers, dishwashers, needs to be domestic. One reason for this is the need to incorporate local requirements in the designs. For instance, refrigerators should be designed to accommodate the big stewpots commonly used in a Turkish household. Another reason is the large logistics costs during importation of bulky items. For example, to import an average size refrigerator from the Far East would add another 10-13% (based on size) to the landed cost.

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Financing: In the durable goods sector the seller finances 85% of its own sales. Since GNP per capita is generally recognized as around USD 3000 per year[1], paying in installments is very important. Dealers get the goods with an average of ten month payment schedule and pass this installment plan to the consumer. Usually, around 50% of receivables are collateralized. This constitutes an enormous open-credit, sometimes much larger than those carried by small banks. In a way, large durable goods shop owners/ dealers act as small business loan providers.

Brand Management: In Turkey there are not different brand names for each distinct type of product of the same producer. Rather, companies prefer to give the same brand name to the entire product line. For example, the brand may be the same both for the Plasma TV and for the radiator or the iron. Therefore, due to the big revenues that can be channeled for the communication and advertising of single brands, durable goods companies have large advertising budgets (Exhibit 16). In fact, four durable good brands including Vestel are among the top ten most recognized brand names in Turkey.

Dealer channel: In Turkey, each durable goods company supports around 1000 dealers, through franchising type agreements. A significant proportion of total sales occurs through the dealer channel. Dealers know their customers and have a role in customer relationship management. Also, these dealers act as credit point institutions since they extend credit to the consumer. Transportation to the household is free.

After-sales service: Companies need to have an extensive after-sales service network in order to serve the consumers and thus gain the market's trust. Having only 25 service points is not enough in an environment where a company like Vestel has over 600 service points throughout Turkey. The service also installs each product for free at a cost to the company.

Early Distribution Initiatives at Vestel Durable Goods

Levent Hatay and Timur Tuncer, Assistant General Manager, joined Vestel Durable Goods Marketing in 1999. Levent Hatay has a PhD in Operations Research. Early on in his career he participated in optimization based projects at American Airlines' Sabre Technologies in the United States. After returning to Turkey, Mr. Hatay worked for Aygaz, a major gas company in Turkey, and Altınyıldız, a major player in the Turkish textile and apparel industry. He led highly successful optimization projects in the distribution of LPG and in production planning of the yarn / dye factory at these companies. Armed with this experience, he joined Vestel Durable Goods Marketing. At that time, Vestel had 1500 dealers but only 100 exclusive shops and Vestel products were also being sold by wholesalers and through different promotions via newspapers etc. A clear marketing approach was not readily apparent. In addition to outlining a medium range marketing plan for the Vestel brand, he worked on restructuring the dealer network (at the end of 2003: 750 showrooms are planned and an introduction of a sub-brand "Regal" for the wholesale market), reviewing the cost structure of the sales organization and restructuring the sales force of Vestel by opening sales offices in various parts of the country. Subsequently, he focused on improvements in two key areas: media planning and distribution. Media planning and distribution are the top two expenses on the firm's books. In the durable goods sector, distribution costs are around 3% of the revenue on the average. Vestel's distribution cost has been lower than this average. Media

[1] GNP per capita for 2002 was USD 2584 according to Turkish Government Statistics Institute

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planning is also important in the Turkish durable goods market due to relatively high advertisement expenses of the companies (Exhibit 16).

SAP

At the time, the company had already implemented SAP, however was making minimal use of it. The software was predominantly used for accounting purposes or used on an ad-hoc basis. In fact, close to a thousand reports were being generated, however most of these were never even looked at. Information was used to document current problems but was not being used for decision making purposes as much as it should. It was clear to the management team that, actually using the functionalities of SAP implied being able to define and measure things. This in turn required discipline in many areas. The sales force was automated by providing laptops and automating end-of-month closing of invoices, reducing the original 5-7 hours per dealer processing time to ten minutes per dealer. A data warehouse was implemented for decision making. The team started taking a closer look at the company's distribution system.

Transportation and Manugistics

The first question posed by the team was: *How do we pay for transportation* at Vestel Durable Goods? The answer received was simple: *washing machine 3 million TL, refrigerator 2 million TL, TV 1.5 million TL* etc (1 $ was 700,000 TL. in 2002). When asked about the differences based on the location of the deliveries, the team was surprised to find out that these prices were uniform across the country, anywhere goods went. What was interesting is that the competitors such as Bosch, Beko, and Arçelik had the same structure. According to Levent Hatay, this led to some unprofessional exchanges in the sector. With competitors this took the form: *For how much do you carry your TVs? Ohh, ours are transported for much cheaper…*With transportation or logistics companies this led to *interactions that would take place between goat vendors in a traditional Anatolian marketplace: Give me 1.5…No 1.75. Brother, seriously it won't work, we will lose a lot* etc… (The numbers refer to million TL and are typically omitted in colloquial exchange). The system was clearly easy to manage, however the company's costs varied by the composition of an order, distribution of orders throughout the country, utilization of the trucks, etc. and none of these differences were being accounted for with the existing system. The most basic question, *is my transportation efficient*, could not be answered empirically.

According to Levent Hatay: *When you are new in a company, it is very important to have information, because if you don't have information, you don't manage by the facts but you manage by what others are telling you to be the facts*. This was the starting point of the project that led to the implementation of Manugistics, a supply chain planning software, at Vestel Durable Goods. The primary objective was not one of reducing transportation costs, but more one of making things more measurable, transparent, and as a result more manageable.

Horoz Logistics

Vestel worked with Horoz Logistics as a third party logistics (3PL) company. Horoz Logistics was selected as a company with good functioning infrastructure and good information technology investments.

Horoz Transportation was founded in 1942. In the 1990s, the local transportation company transformed to be a third-party-logistics company that provides integrated services in international freight forwarding, custom clearance,

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warehousing, cargo and parcel distribution etc with an ISO 9002 certification. In 2000, Horoz Logistics formed a partnership with the French company SDV. With its 1880 employees and an international fleet of 200 vehicles, Horoz Logistics' 2002 revenue was $113 million where the domestic truck transportation constituted 23% of the revenue.

The logistics outsourcing agreement between Vestel and Horoz was probably the largest one in monetary value in Turkey. Horoz Logistics is solely responsible for inbound international logistics of Vestel from 69 different countries with air-, sea-, and truck-shipping and custom clearance. Horoz manages the distribution from the Vestel factory in Manisa to all the dealers in Turkey. Horoz also coordinates Vestel's exports from Manisa to 144 different countries with containers. Vestel is Horoz's largest customer and a majority of Horoz's revenues comes from this agreement.

Horoz managed its own truck fleet, but also relied extensively on truck cooperatives within which individual truck owners operated. There were some legal obligations associated with using the trucks of truck cooperatives, and in some cases the logistics companies did not have another choice. The truck cooperatives can be compared to labor unions in other settings. The relationship between Horoz and the truck cooperatives was to a large extent transparent to Vestel.

Pricing with Horoz Logistics: the old system

Before the implementation of the new system, with the flat price per item pricing scheme, the only parameter necessary for billing was the number and type of items shipped. Due to manual verification processes and bad documentation on both sides, it was not uncommon for Vestel Durable Goods and Horoz Logistics to disagree at the end of a month, and billing sometimes took 15 days to complete. Once again, the major problem of the old system was that it could not be measured properly. Furthermore, the pricing scheme made it hard for Horoz to account for its own costs and profitability. Agreements with truck owners from the truck cooperatives were typically done on a per truck basis and varied by region being traveled to. This market was highly competitive so a typical truck cost was marginally higher than the diesel costs associated with the trip. The cargo distribution network of Horoz Logistics is depicted in Exhibit 15.

The Vestel Distribution Network

Most of the production occurs in a plant in Manisa. Imported goods are also received there. Until 1999, the company had four warehouses, serving the dealers and outlets in different regions of the country (Exhibit 3). Distribution is performed by Horoz Logistics. With the flat price per item pricing scheme given by the third-party-logistics (3PL) company, it soon became clear to everyone that there was no need to keep four warehouses. This led to an initiative of warehouse consolidation, whereby the distribution network took its current form with two warehouses. Currently, Vestel has two warehouses one in Manisa next to the production facilities and another one in Istanbul. The total warehouse area is 25,500 m^2. Today, the company does not consider its network as final, and is reevaluating new warehouse locations in light of changes that have occurred in the mean time.

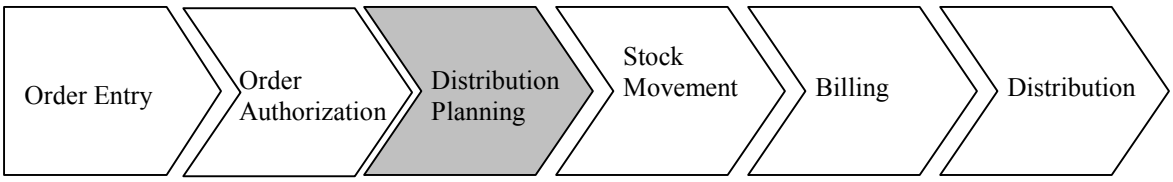
The sales in a given month differ substantially between regions in Turkey (Exhibit 6). Fluctuations in weekly shipment patterns further complicate production planning and distribution for Vestel (Exhibit 5). These patterns are common for all the other players in the market. Moreover, the difference between the loads of outbound and inbound traffic increases the transportation costs, since truck owners reflect this on the prices that they charged.

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A New Planning System: Manugistics Transportation Management

Given the objective of a better measurable system, Vestel decided to implement Manugistics' Networks Transport Management (MTM) module as the next improvement effort for the distribution system in 2000. This package was chosen based on service options made available in Turkey by the various SCP providers and subsequent to a negotiation on price. Vestel Durable Goods Marketing was the first company in Turkey to implement such a transportation planning system, and remained the only company in 2003.

The distribution planning program is run daily to schedule deliveries to Vestel's customers. The planning process is a part of the order fulfillment process. The flow of the order fulfillment process is as follows:



The order fulfillment process

The transportation planning performed by MTM only constitutes one part of the order fulfillment process. This process starts with order entry by the dealers. This can be done either online directly into SAP, offline using MS Access, or via the web. All orders for which a delivery notice has not been generated are in pending status.

Generation of the delivery notice starts the billing process. These notices can be generated manually any time of the day (as was done before MTM), or are generated automatically between 12:00 and 13:00 every day. For the manual order generation, a list of orders and the amount of usable stock on hand that is displayed next to this list is used. Available stock is assigned to orders, and as these assignments are made, the stock level is adjusted. The current system sorts the orders by date, and within each date, sorts them by prioritized sales outlets. The credit check is automatically performed upfront before the orders are generated.

Once orders are generated in this fashion, the data is fed to the distribution planning module. The software determines a near-optimal distribution plan for that day, given the deliveries that need to be made. The plan gives the assignment of goods to trucks, truck routes and transfer stations. The decision whether the goods should be sent via cargo or by truck is also made. Within the distribution planning module, some deliveries may be held due to a truck utilization constraint. For all deliveries that are not held for a given day, information is retransmitted to SAP, where the necessary documents to physically move goods from stock are generated, stock levels are adjusted, and bills are printed at around 17:00, and sent to the warehouse. The truck and van needs identified by the planning software are communicated to the logistics company along with the planned routes to be used for those deliveries. This process is repeated six days a week. The order fulfillment process flowchart is summarized in Exhibit 17.

According to Mr. Oktay Esendemir, Planning Manager, this process is a good example of how Manugistics helped the company to further its use of SAP functionalities, and automate key processes. In the order fulfillment process, stock allocations were rationalized. The automatic credit check reduced the number of eligible dealers for delivery, thereby

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enhancing delivery volumes and truck utilizations. Furthermore, the company could now tell precisely which dealer was shipped which items, and as a result managing the dealer relationships became easier to do.

MTM capabilities

MTM is a transportation optimization software program, which provides the optimal route and truck planning for daily-prepared deliveries (Exhibit 18). The inputs to the system are locations of Vestel's warehouses, transfer stations, and its customers; customer orders, transportation modes, and associated costs. The optimization program uses these inputs and finds a solution within the constraints imposed by the management to minimize the total transportation costs. The route and truck planning is made according to the inputs and the constraints.

There are 3 different location types in MTM: warehouse, transfer station, customer. All the locations have zip codes created specifically for MTM. These codes are different for each province. Some big provinces are divided into two or more regions. The distances between each two zip codes are put in a network table. The distance between two points located in the same zip code is set to be 3 km.

Vestel Durable Goods Marketing Inc. has two warehouses, one in Manisa and the other in Istanbul. While the warehouse in Istanbul is serving customers in Istanbul, Sakarya, Kocaeli, Yalova, Tekirdağ, Kırklareli, Edirne and the north of Çanakkale the warehouse in Manisa works for the remaining cities in Turkey.

There are 9 regions throughout Turkey and the total number of transfer stations in these regions is 19. The logistics company owns and operates these stations. Exhibit 15 shows the transfer stations used by Horoz Logistics.

The volume information for each product is provided as an input into the system. For example, the volume of a particular model of washing machine (800C) is 332 dm^3 while a turbo oven (3310T) occupies 447 dm^3.

Three different size trucks can be used for transportation in addition to a direct cargo alternative. The costs of using each alternative are set in the system. 10-wheel (32,000 dm^3) or 8-wheel (25,600 dm^3) trucks are used for the transportation to transfer station from the warehouses. Small trucks (14,000 dm^3) then make the deliveries from the transfer stations to the customers. There is also a direct cargo alternative from the warehouse in Manisa. Dealers with high volume demands can have direct deliveries with large trucks. MTM selects the direct cargo option based on transportation costs. Truck utilizations constitute an important criterion for deciding on delivery mode.

The management uses two policies related to efficiency and customer service. The first policy is related to truck utilization. A truck has to be at least 65% full in order to depart for its destination. Otherwise it waits until this rate is achieved. The maximum waiting time is the other policy related to customer service. This waiting time is restricted to be at most 3 days to provide a good service to distributors. After 3 days even if a truck is not 65% full it will leave the warehouse either by truck or by cargo, whichever is more efficient.

As a result, once the planning is made in MTM, transportation can be in two states:

- En-route: The truck and the route have been planned, transportation can take place

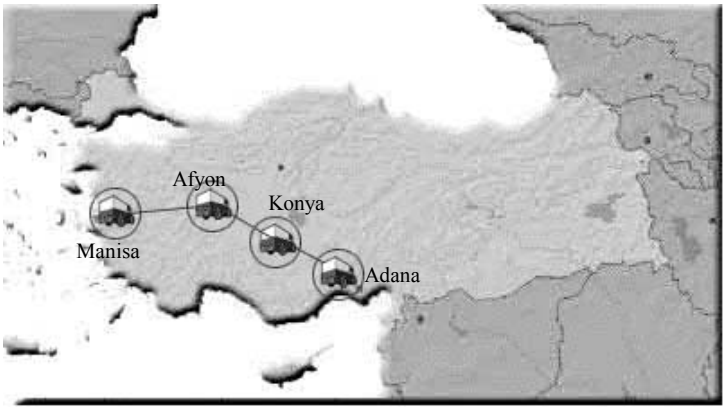
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- Routed: The truck and the route have been planned, transportation can not take place, and the deliveries will wait until the next day

The deliveries planned as cargo are, by default, let in the "routed" status. A program that is run after the planning phase searches the entire cargo traffic list and switches the status of cargo deliveries waiting for more than 2 days to en-route and enables their deliveries on that day. By letting cargo deliveries wait in the routed status, the system allows for the possibility of changing these to regular truck deliveries with the arrival of additional similar demand, thereby reducing costs.

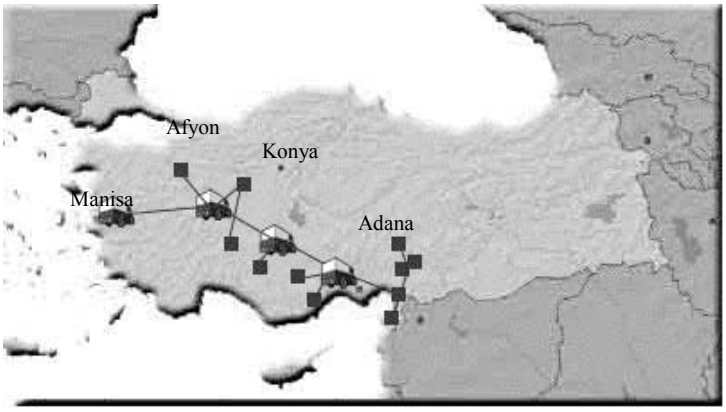
MTM does not optimize truck loading. Since MTM does not plan inside the truck a loading problem may occur. Given the difference in shape of the various goods being transported, not all items planned by MTM may be loaded on a truck due to space constraints. As a result, volumes were increased to enable the feasibility of the plans generated by the software. While truck load optimization would be feasible for simple deliveries between two points, the Vestel distribution problem is significantly more complex due to routes that have multiple drop-off points. As a result, the planning objective is not to find the loading that maximizes truck utilization, but rather the loading that allows for the best unloading of trucks without having to load and unload different items at the various drop-off points.

The planning approach works in three phases: loading of trucks, routing to transfer stations, and routing from transfer stations. In the example below, MTM routes a large truck from Manisa first to Afyon, and then from Afyon to Konya, and finally from Konya to Adana transfer stations.



In this phase, MTM achieves 95% to 100% utilization of trucks. In the second phase, transfers to surrounding locations are planned. For example, small trucks are scheduled from Afyon to Afyon, Isparta, and Kütahya dealers, from Konya to Konya dealers, and from Adana to Adana, Kahramanmaraş, Hatay, and İçel dealers.

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Let us consider a typical day at Vestel. Orders from various customers are collected. One of these orders is from Köprü Inc., a dealer in Çankaya, Ankara. They ordered 2 Turbo ovens (models 3310T and 3400T) and 1 Washing Machine (Model 800C). The distribution plan first puts these goods in a large truck (truck no TL120) with other orders from various dealers. Since on this date, there are orders from other dealers from the Ankara region, the truck is sent directly to Ankara from Manisa. At the transfer station in Ankara, it is scheduled to be put in a small size truck (truck no TL102) with orders from two other dealers in the region. The plan also shows that this truck delivers the goods of Köprü Inc. at the fourth stop of its trip. What is surprising in this plan is that the truck utilizations are 100% and 99.58%. The routing tables generated as the output of MTM for this example can be seen in Exhibit 14.

In 2002, Vestel scheduled on average 125 trucks every day and delivered 49,000 products to 1000 different locations every month by using this planning system (Exhibit 12).

Pricing with Horoz Logistics: the new system

The biggest impact of the MTM software was on the logistics company. Vestel's use of MTM was unique, in the sense that the planning was done by them for the 3PL company. Before MTM, Horoz Logistics had routes that were pretty much fixed, based on the experience of the drivers and planners in the company. To them, the idea of following a route that was generated by a software package was new, and in some cases the routes generated seemed wrong. This led to special meetings early on, where both parties convinced themselves that the routes generated by MTM were indeed making sense.

Once MTM was implemented and started running on a daily basis, it became clear to Timur Tuncer and his team, that a smarter pricing scheme with the logistics company could be designed. In particular, the new system allowed both parties to monitor how many trucks of each type were being used, and how far each of them was traveling. Both this information and the details about which good was being delivered in which truck via what route were being kept in Manugistics. This made it possible to switch to a new price per truck per distance scheme.

The transportation prices were determined by the truck cooperative and given to the logistics firm according to two criteria: the type of the truck and the destination region. Based on this, Horoz Logistics determines a price for each truck per km. The destination region is an important criterion, since for some regions like the east of Turkey the truck

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will return empty most of the time, due to the imbalance in economic flows between the west and the east. The truck cooperatives try to take this into account when determining their regional prices. Six to seven different regions were considered when determining prices. Manugistics was only responsible for planning the one-way route and did not consider the return. The return traffic consisted only of return items, as such volumes were low and the cargo alternative was typically preferred.

The price thus quoted consists of two parts: fixed costs and the cost of the diesel. Whenever there is a change in the diesel price the shipment price is updated immediately, otherwise the updates are made once every 6 months. The diesel price makes up 29% of the total price.

For cargo deliveries, per volume prices are set for four different distances (0-200, 201-600, 601-900, and 900- km). MTM calculates the cargo prices automatically and generates a payment to Horoz Logistics once an item is sent via cargo.

This new pricing scheme was a radical change, and allowed both parties to account better for their expenses. For Horoz Logistics, the pricing was more natural, given their own pricing scheme with the truck cooperative. For Vestel, the scheme allowed to reduce transportation costs significantly compared to the old system. This can be seen in Exhibit 9, where volume information for each type of item was used to convert the new price to an equivalent price per item in order to enable a comparison.

Implementation Issues for Vestel

The results obtained from the implementation of Manugistics were phenomenal. Truck utilizations went up (Exhibit 13) while the transportation costs decreased between 1999-2003 (Exhibit 9). In 2002, transportation costs were decreased by 46% despite the increase in diesel prices and increase in Consumer Price Index (Exhibit 8). The unit cost of transportation per item went down in some cases by as much as 75% (Exhibit 11).

In addition to the new planning system, a number of other factors were also instrumental in achieving high utilization rates. First, the number of orders entered manually into the system decreased. Increase in pre-paid orders helped to achieve a more even distribution of the orders within a month (Exhibit 5). Moreover, the total volume also increased in 2003 (Exhibit 13).

The impact of the software was felt throughout the supply chain. According to the Planning Manager: *we have found comfort and convenience*. Within Vestel planning, everyone was happy with the new system, however there was resistance elsewhere. According to Timur Tuncer: *if it weren't for the general manager's strong will and leadership, this project would never have happened*. The sales department that was the liaison between the company and the dealers was uneasy about the implications of the new system on their relationship with their key customers. The dealers expressed their mistrust of the new system by calling all delivery delays, which used to happen in the old system as well, as *Manugistics delays*. It was claimed that the software was holding back the deliveries. The unhappy dealers also made the sales organization unhappy. The sales people in turn used the software as a scapegoat from time to time. For

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example when a dealer failed a credit check, rather than telling their customers this had happened, they preferred to say that Manugistics had failed in generating their deliveries.

However as time passed it became clear to everyone that the new system was better over all. The dealers could track their orders via the internet, and all goods that left the plant had dates and barcodes fixed on them so that delivery lead times became transparent. The reduction in transportation costs could be reflected on end customer prices, which improved the dealers' competitive position. In the annual dealers meeting, management was able to address all dealer questions pertaining to transportation with precise numbers and facts. Since everything was measurable in the new system, Levent Hatay could show the actual data related to on time deliveries at the last meeting of the dealers instead of arguing on anecdotal complaints. In 2002, on time delivery ratio reached 84%.This removed transportation related skepticism from the dealers' minds.

Implementation Issues for Horoz Logistics

From the logistics company's perspective, implementation of the new system introduced some challenges at the early stages. By working together with Vestel, both parties found a solution to each of the problems.

Initial routes that were planned with multiple drop-off locations introduced difficulties for Horoz Logistics at the beginning. When loading and unloading times were not taken into account by the route planning procedure, MTM produced routes that could not be finished within one day. This required that the truck be driven to the nearest transfer station for the night, goods on it unloaded to a safe location, and then reloaded onto the truck the next morning to resume the route. This created additional costs for the logistics company. Once the issue was brought to Vestel's management, constraints were added to the model in Manugistics to limit the number of drop-off locations in the routes produced.

Horoz Logistics were also affected by the fluctuation in volumes of goods shipped through transfer stations. A transfer station run by Horoz Logistics could experience two days where in the first day there would be no trucks that utilize the transfer station, and in the subsequent day six large trucks transferring goods onto twenty vans could be required. Given the direct encounter between the end customer where deliveries were made and the van drivers, this implied that not any van (and as a result van driver) could be used in a transfer station. These drivers needed to have the minimum skills required for loading, unloading, invoicing, and basic customer relationships, implying that Horoz Logistics could not rely on daily part-time arrangements. As a result, the company needed excess capacity at each transfer station with important fixed costs attached. Vestel helped the logistics company with this problem by using simulation capabilities that MTM provided in order to determine which transfer stations should be kept and which closed based on the underlying utilizations and resulting economics. Vestel also started transmitting the routing plans immediately when they were created to help Horoz to manage its capacity in a better way. By learning what the plan is for a transfer station and how many truck are needed, one- to two-days in advance helped Horoz to manage its transfer stations more effectively.

One of the difficulties with MTM, according to Horoz Logistics CEO Taner Horoz, was the geographic information system being used within the system. Supplied by a local software company, INFOTECH, this provided the distance information that was used by MTM for the route optimization. While the distance information was accurate for between

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city long distances, shorter distances, especially those within cities were not very accurate in some instances, thus leading to some errors in the subsequent optimization. Taner Horoz viewed the lack of reliable input data in the form of distance matrices as one of the major problems for implementing such a package in Turkey. Management at Vestel believed that the problematic measurements were outliers and were determined to remedy this problem by adjusting certain disputed distances and payments made to the logistics company based on these. Both parties initiated a project to study the issue and correct possible inaccurate distances that are used by MTM in December 2003.

The CEO Taner Horoz stated: *Ideally, a logistics firm would like to establish their own distribution plan. We are capable of providing our customers economies of scale, and that is one of the key reasons for our existence. The current system with Vestel is designed to optimize one customer's problem. For the micro distribution part of the problem, we use opportunities to consolidate some shipments with those of other clients. I have highlighted some problems that we experienced with the system. However we have a long term relationship with Vestel and believe that whatever problems there are in such an initiative, both parties will do their best to remedy any problems that any one of the parties is experiencing in this relationship. We view them as partners and have a trust in the strength of our relationship. Our ideal would be to perform the distribution planning jointly with Vestel.*

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Next Step

As Levent Hatay approached his office, he thought about the next steps to be discussed in the meeting. Could the distribution system of the company be further improved? He thought: *We can use Manugistics as a simulation tool to reassess our distribution network and determine the ideal design in terms of the number of warehouses. Mr. Arda Atalay, our IT manager is already working on this, along with a simulation to assess the use of 18-wheel trucks. Another thing we are working on is a more flexible system, where the transfer stations are not fixed but can be opened or closed dynamically within the optimization. This would require collaboration from Horoz, but by relaxing these constraints in the optimization, we could gain a lot of mileage. We have the tool, so we can continuously improve our current system.*

But what about something more radical? Even when we have done most of what can be done internally, there will still be room for improvement in the sector. What if we collaborated with some of our competitors on transportation? This would certainly fill-up our vans better. What other benefits could we expect? Would we share our planning with them? American Airlines shares its fleet management system with US Air and Lufthansa. They are also competitors. Actually there is room for collaboration in our cargo problem as well. Currently, we haven't imposed a constraint on cargo capacity within Manugistics. We want to see what is optimal this way. But the cargo requirements generated by the program exceed Horoz Logistics' cargo capacity. Why shouldn't I turn to other cargo service providers if that is the case? Cargo capacity is perishable anyway. What would this mean in terms of our relationship? Yes, this should be the theme of today's meeting: collaboration and relationships.

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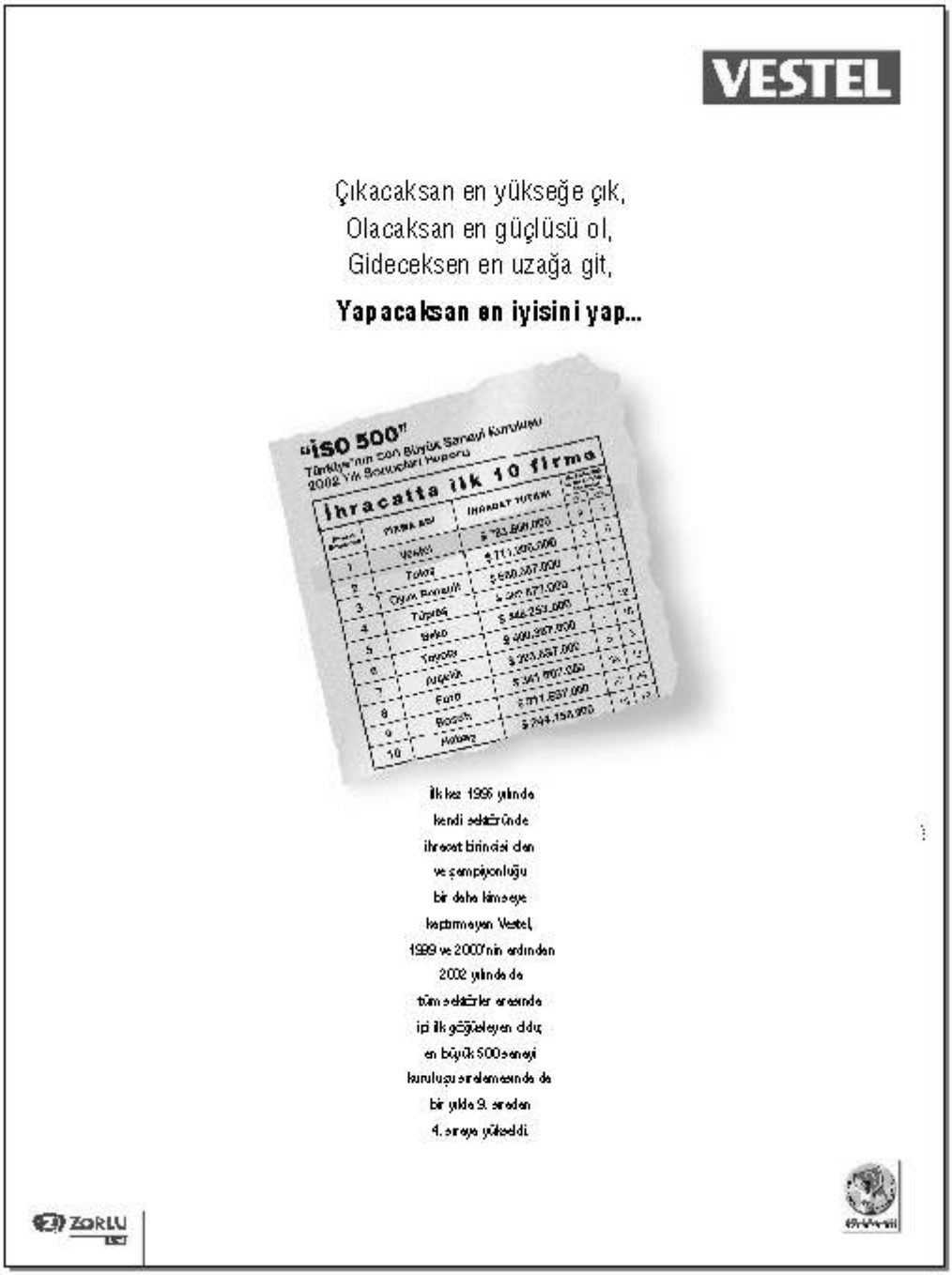
Exhibit 1. Revenues of Vestel Group Companies

	2000 ($)	2001 ($)	2002 ($)	2003* ($)
Vestel Foreign Trade	634,482,218	664,501,915	1,022,335,450	1,388,446,273
Vestel Electronics	799,597,995	764,869,391	966,538,152	1,096,273,870
Vestel White Goods	66,743,973	75,519,884	132,150,216	218,576,272
Vestel Communications	143,539,369	128,592,276	181,066,292	281,865,405
Vestel Durable Goods Marketing	238,079,128	122,777,528	172,629,100	318,940,992

*estimated figure based on 2003 Jan-Aug figures

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Exhibit 2. Vestel ads



(*Vestel ranked number one among all the Turkish exporters in 2002*)

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(*Turkey's first air conditioner's with Ion Technology: BioPlus*)

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(*Vestel's new TFT-LCD Television*)

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Exhibit 3. Turkey and the Distribution Network of Vestel



Exhibit 4. The Number of Units Shipped: Annually and Monthly

	2000	**2001**	**2002**	**2003***
Annual	900,000	518,867**	592,652	1,007,701
Monthly	75,000	43,239	49,387	83,975

*estimated figure based on 2003 Jan-Aug figures

** In 2001, due to the financial crisis in Turkey, the durable goods market reduced by 48%

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Exhibit 5. Weekly Distribution of the Monthly Revenue and Truck Utilizations

		First Week	Second Week	Third Week	Fourth Week
Weekly Distribution*	**2003 January-June**	19.5%	20.8%	24.3%	35.4%
	2002 January-June	10.28%	17.17%	19.91%	52.65%
	2002 January-December	9.92%	18.42%	19.62%	52.04%
Cumulative Truck Utilization	**2003 January-June**	93%	94%	71%	90%
	2002 January-June	41%	44%	53%	79%
	2002 January-December	78%	72%	75%	78%

*Percentage of the monthly revenue distributed within the week

Exhibit 6. Geographical Distribution of Total Revenue (March 2003)

Region	Share in Total Revenue
Istanbul (European)	21%
Adana	14%
Chain Markets and wholesale	12%
Istanbul (Asian)	11%
İzmir	10%
Ankara	7%
Central Anatolia	5%
Marmara	5%
Samsun	5%
Aegean	4%
Antalya	4%
Erzurum	2%

Exhibit 7. 2003 January-June Television Total Production and Market Shares

	DOMESTIC SALES		EXPORT SALES		TOTAL SALES	
BRAND	**Units**	**Market share**	**Units**	**Market Share**	**Units**	**Market Share**
VESTEL	266,493	37.3	2,916,984	50.88	3,183,477.0	49.4
BEKO	244,120	34.2	1,924,586	33.57	2,168,706.0	33.6
TELRA	141,296	19.8	888,819	15.51	1,030,115.0	16.0
İMPER	22,994	3.2	2,401	0.04	25,395.0	0.4
PHILIPS	39,879	5.6	0	0	39,879.0	0.6
TOTAL	**714,782**	**100.0**	**5,732,790**	**100**	**6,447,572.0**	**100.0**

(Source: ECID reports 2003)

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Exhibit 8. Consumer Price Index and USD Exchange Rate Change from 1999 to 2003

	1999	2000	2001	2002	2003*
***Index Inflation* (TL)**	100.00	151.43	244.76	367.40	461.50**
Index $ Exchange Rate	100.00	124.38	266.54	302.63	263.23***

* estimated figure based on 2003 Jan-June figures

** 100 TL of 1999 is equivalent to 461.5 TL of 2003 because of the high inflation rate during 1999-2003

*** $ exchange rate increased 263.23% during 1999-200.

Exhibit 9. Decrease in Total Transportation Cost from 1999 to 2003

	1999	2000	2001	2002	2003*
Index Trans. Cost /Sales Revenue	100.00	119.92	96.69	84.08	81.32
Index. Trans. Cost /Cost of Goods Sold	100.00	118.21	98.84	87.91	80.67
Index of TL/dm^3 transportation	100.00	109.82	124.61	158.93	163.12**

* estimated figure based on 2003 Jan-June figures

Exhibit 10. The ratio of the unit transportation cost to the unit sales price.

	2000	2001	2002
Refrigerator	4.26%	4.02%	2.62%
TV	2.46%	1.07%	0.89%
Washing machine	2.72%	1.36%	1.07%
Dish Washer	2.94%	1.45%	1.06%
Stove	2.98%	2.22%	1.30%

Exhibit 11. The unit transportation cost decrease between 1999-2002

Products	% Change In USD
TV	-42.92%
Washing Machine	-47.36%
Refrigerator	-12.92%
Dish Washer	-44.73%
Mini music player (portable)	-68.62%
Midi music player	-41.50%
Micro music player	-28.64%
Small home appliances	-51.25%
Receiver	-66.41%
Dish Antenna	-33.52%
TV rack	-62.49%
Mini Refrigerator	-48.97%
Carpet Washing machine	-29.31%
Air conditioner (split)	-76.69%
Air conditioner (window)	-64.48%
Computer	-66.20%
Aspirator	-52.06%
Oven	-60.77%
Stove	-34.78%
Flash heater	-45.10%
Vacuum cleaner	-71.20%

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Exhibit 12. 2002-2003 Transportation Figures

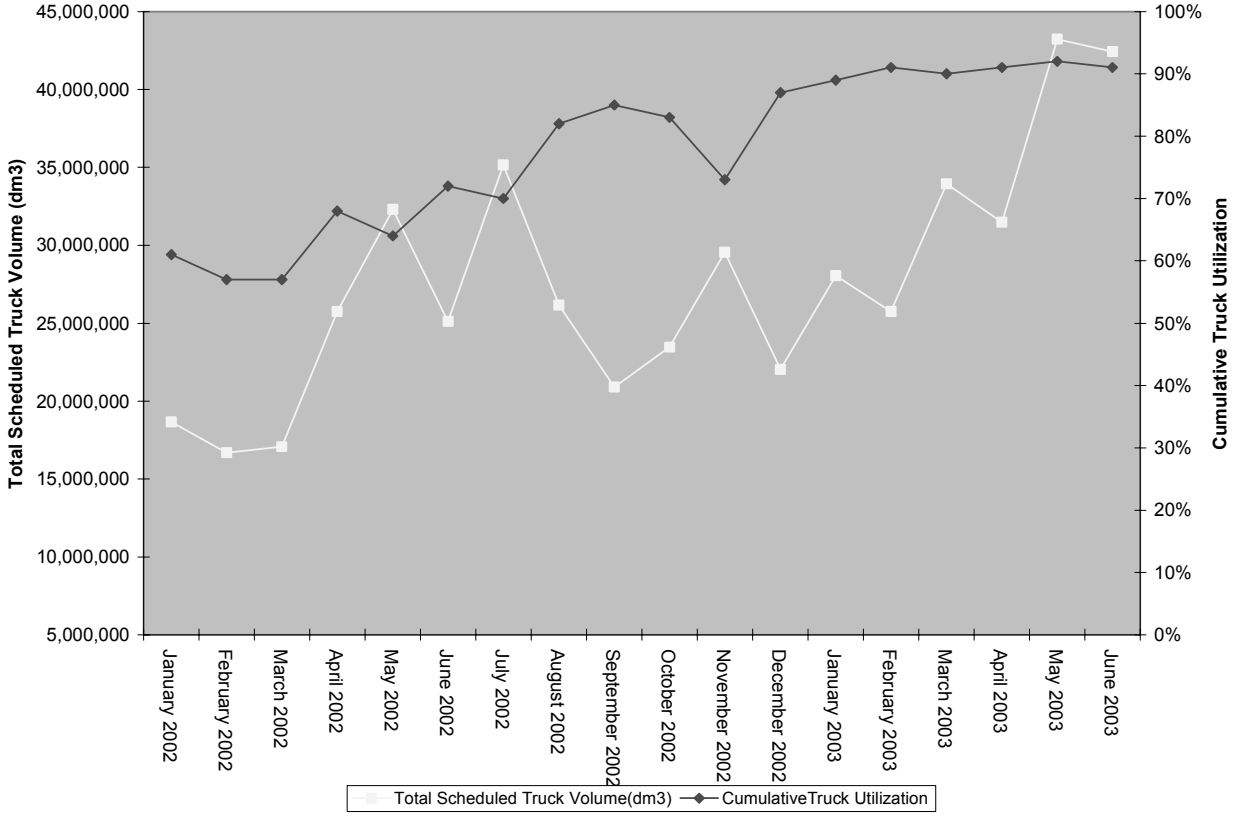
		Amount*	Total Transported Volume(dm^3)	Total Scheduled Truck Volume (dm^3)	Cumulative Truck Utilization**	The Number Of Trucks
2002	**January**	41,153	11,389,673	18,667,200	61%	Small Truck 623 10-wheel 187 8-wheel 145
	February	43,160	9,544,683	16,691,200	57%	Small Truck 640 10-wheel 148 8-wheel 107
	March	35,591	9,712,160	17,062,400	57%	Small Truck 640 10-wheel 130 8-wheel 144
	April	46,284	17,525,463	25,747,200	68%	Small Truck 704 10-wheel 335 8-wheel 191
	May	58,658	20,827,323	32,291,200	64%	Small Truck 974 10-wheel 406 8-wheel 206
	June	64,319	18,020,355	25,102,400	72%	Small Truck 861 10-wheel 309 8-wheel 110
	July	60,552	24,589,658	35,147,200	70%	Small Truck 1171 10-wheel 445 8-wheel 158
	August	46,983	21,504,588	26,148,800	82%	Small Truck 803 10-wheel 371 8-wheel 106
	September	43,418	17,827,079	20,894,731	85%	Small Truck 622 10-wheel 221 8-wheel 80
	October	52,533	19,580,300	26,940,860	73%	Small Truck 778 10-wheel 279 8-wheel 130
	November	69,612	21,697,932	32,733,792	66%	Small Truck 981 10-wheel 401 8-wheel 101
	December	68,257	19,203,152	25,111,986	76%	Small Truck 802 10-wheel 258 8-wheel 87
2003	**January**	77,063	22,833,713	28,046,400	89%	Small Truck 969 10-wheel 382 8-wheel 73
	February	82,877	24,038,651	25,745,600	91%	Small Truck 787 10-wheel 444 8-wheel 8
	March	104,717	29,496,274	33,944,000	90%	Small Truck 951 10-wheel 568 8-wheel 81
	April	115,406	28,067,089	31,480,000	95%	Small Truck 851 10-wheel 560 8-wheel 51
	May	158,242	35,513,913.76	43,228,800	93%	Small Truck 1326 10-wheel 675 8-wheel 99
	June	154,923	37,430,499.93	42,427,200	90%	Small Truck 1289 10-wheel 673 8-wheel 91

* Partial shipment + complete truck transportation + Inter-warehouse transportation + Service centers

** Cumulative Truck Utilization =Total Transported Volume (dm^3)/ Total Scheduled Truck Volume (dm^3)

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Exhibit 13. Changes in Total Scheduled Truck Capacity and Cumulative Truck Utilization



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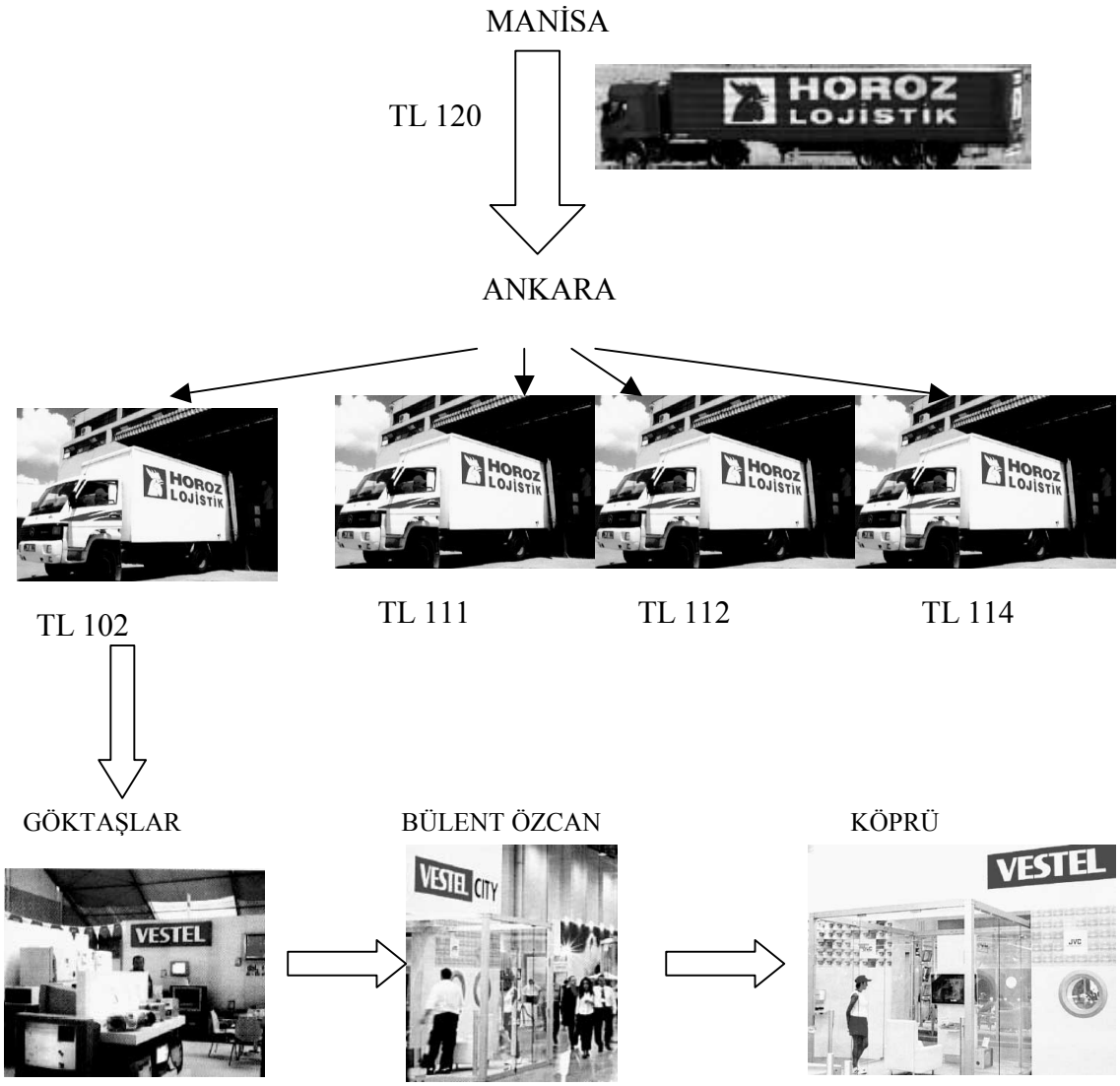
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Exhibit 14. Example: Routing Tables Generated by MTM

This example shows how the orders of various distributors in Ankara area are planned to delivered by the MTM system. The proposed plan sends one 10-tire large truck (truck number TL102) from Manisa to Ankara. Then four smaller trucks (trucks TL102, TL111, TL112, TL114) are loaded in Ankara. Each truck delivers the orders of various distributors. For example, truck TL102 delivers the orders from three distributors.

The generated forms include all the details about trucks, orders, customers, and loading-unloading operations. Schematic representation of this plan is shown below:





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Vestel Durable Goods Marketing

Loading Information Report

Truck No: TL120 **DPT:** MAN00151030M **Truck Type:** 10-TIRE
Plan: PLAN1 **Truck Capacity:** 32000
Departure 2204 - / MANISA **Total Order Capacity:** 32000
Arrival XANKARA – ANKARA TRANSFER / ANKARA **Remaining Km Capacity**: 0
Occupancy Rate: 100.00%
Loading Date: 11/04/03 **Status:** D

Stop 1
L / U : Load **Location:** 2204 – MANISA WAREHOUSE (MANISA)

Delivery No	Row No	Material No & Name	Quantity	Volume (dm^3)
0220034266	01	20220794 REGAL DVD 3000 DVD PLAYER	1	20
0311043786	01	20220624 AV32X25 JVC TV RACK (RK-32C32B1SW)	2	186
0311044535	04	20220488 ZX 72 TXT STEREO	40	12,928
0311044542	01	20220488 ZX 72 TXT STEREO	8	2,586
	02	20220752 ONYX 70 TXT S	8	2,440
0311044549	01	20220488 ZX 72 TXT STEREO	30	9,696
	10	20220486 ZX 51 TXT	20	4,144

Stop 2
L/U: Unload **Location:** XANKARA – ANKARA TRANSFER (ANKARA)

Delivery No	Row No	Material No & Name	Quantity	Volume (dm^3)
0220034266	01	20220794 REGAL DVD 3000 DVD PLAYER	1	20
0311043786	01	20220624 AV32X25 JVC TV RACK (RK-32C32B1SW)	2	186
0311044535	04	20220488 ZX 72 TXT STEREO	40	12,928
0311044542	01	20220488 ZX 72 TXT STEREO	8	2,586
	02	20220752 ONYX 70 TXT S	8	2,440
0311044549	01	20220488 ZX 72 TXT STEREO	30	9,696
	10	20220486 ZX 51 TXT	20	4,144

Related Transportation Steps

Truck No	Departure Point
TL102	XANKARA – ANKARA – 00000150896M
TL111	XANKARA – ANKARA – 00000250975M
TL112	XANKARA – ANKARA – 00000150977M
TL114	XANKARA – ANKARA – 00000150992M

! shows the deliveries not loaded after 3 days

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Vestel Durable Goods Marketing

Loading Information Report

Truck No: TL102 **DPT:** 000001500896M **Truck Type:** SMALL TRUCK
Plan: PLAN1 **Truck Capacity:** 14400
Departure XANKARA - / ANKARA **Total Order Capacity:** 14340
Arrival 20300678 – KÖPRÜ DAY.TÜK.MAL CANKAYA1 / ANKARA **Remaining Km Capacity**: 60
Occupancy Rate: 99.58
Loading Date: 11/04/03 **Status:** D

Stop 1
L / U : Load **Location:** XANKARA – ANKARA TRANSFER (ANKARA)

Delivery No	Row No	Material No & Name	Quantity	Volume (dm^3)
0311043786	01	20220624 AV32X25 JVC TV RACK (RK-32C32B1SW)	2	186
0311044535	04	20220488 ZX 72 TXT STEREO	40	12,928
0311044569	02	20210434 CV 3400 T TURBO OVEN	1	447
	03	20210435 CV 3310 T TURBO OVEN	1	447
	04	20210502 WMU 800C WAHSMACHİNE	1	332

Stop 2
L / U : Unload **Location:** 20300015 – GÖKTAŞLAR İÇ TİC. TURİZM / ANKARA)

Delivery No	Row No	Material No & Name	Quantity	Volume (dm^3)
0311043786	01	20220624 AV32X25 JVC TV RACK (RK-32C32B1SW)	2	186

Stop 3
L / U : Unload **Location:** 20300034 – BÜLENT ÖZCAN (CANKAYA1 / ANKARA)

Delivery No	Row No	Material No & Name	Quantity	Volume (dm^3)
0311044535	04	20220488 ZX 72 TXT STEREO	40	12,928

Stop 4
L / U : Unload **Location:** 20300678 – KÖPRÜ DAY.TÜK.MAL (CANKAYA1 / ANKARA)

Delivery No	Row No	Material No & Name	Quantity	Volume (dm^3)
0311044569	02	20210434 CV 3400 T TURBO OVEN	1	447
	03	20210435 CV 3310 T TURBO OVEN	1	447
	04	20210502 WMU 800C WAHSMACHİNE	1	332

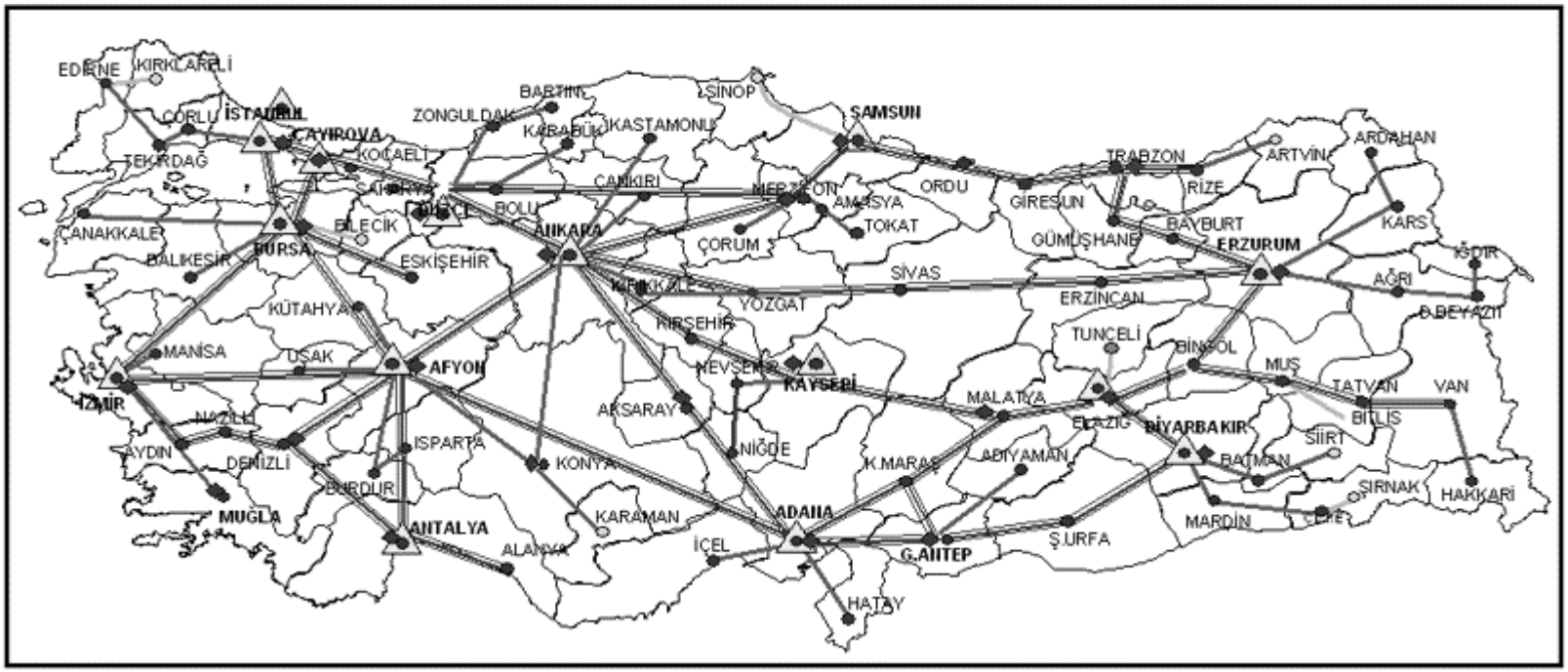
Related Transportation Steps

Truck No	Departure Point
TL116	2204 – MANİSA – MAN00151003M
TL120	2204 – MANİSA – MAN00151003M

! shows the deliveries not loaded after 3 days

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Exhibit 15. Horoz Logistics Distribution Network



Regional Centers
Transfer centers
Offices
Main routes
Ring routes
Mobile delivery routes

15 Regional Centers
22 Transfer centers
236 Offices

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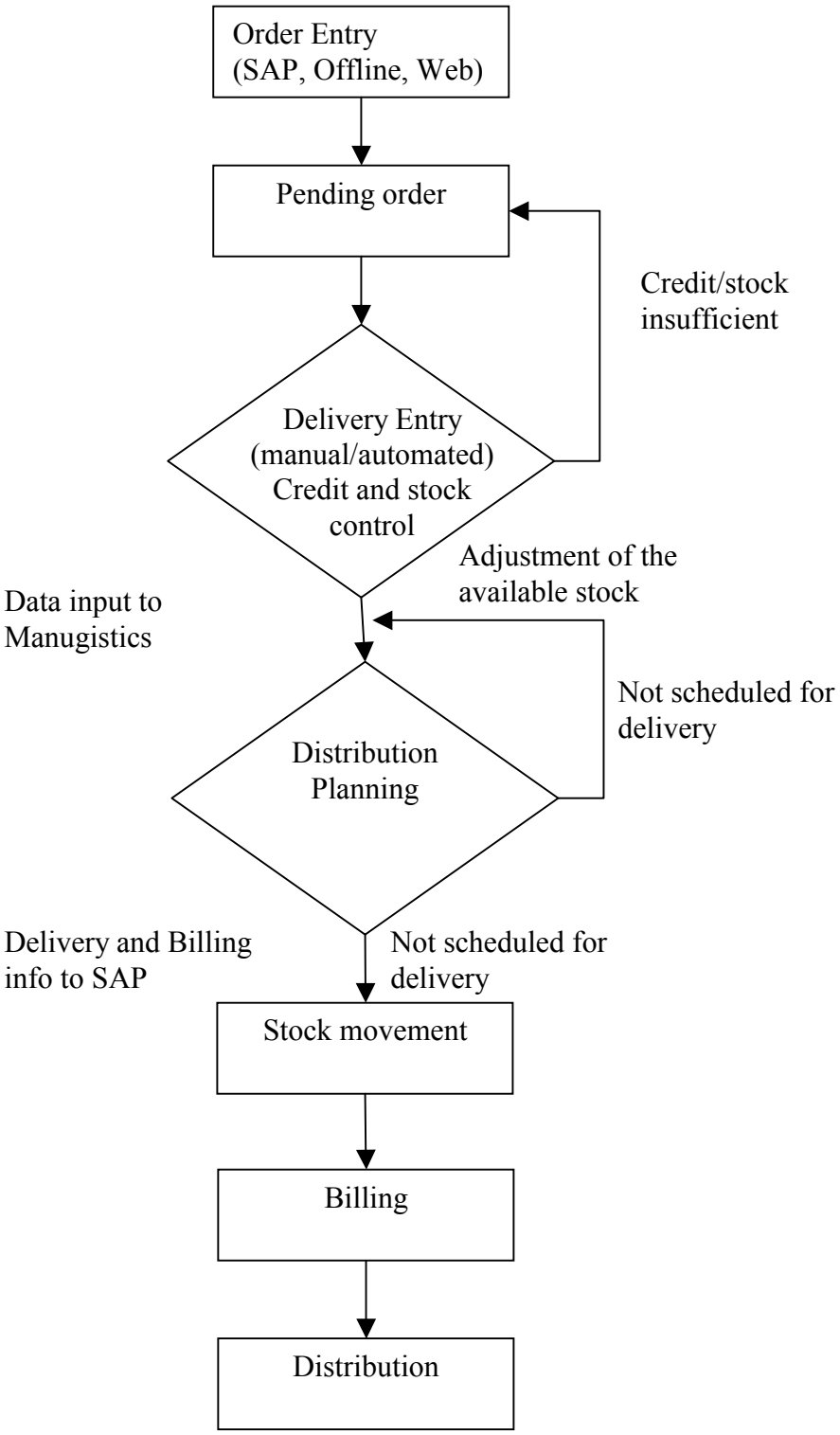
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Exhibit 16. 2002 General Advertisement Expenses by Channel (Ratecard-$)

2002	TV		NEWSPAPER		MAGAZINE		RADIO		OUTDOOR		CINEMA		TOTAL AMOUNT-$
BRAND	SECONDS	AMOUNT-$	COLUMN X CM	AMOUNT-$	PAGE	AMOUNT-$	SECONDS	AMOUNT-$	QUANTITY	AMOUNT-$	QUANTITY	AMOUNT-$	
VESTEL	101,216	16,966,370	56,568	6,883,842	69	386,664	425,569	6,251,177	2,400	256,642			30,744,694
VESTEL JVC			5,252	751,767	38	208,595							960,362
ARCELIK	272,964	39,641,904	65,084	9,192,563	57	248,627	355,724	4,925,900	4,020	417,211	596	4,426,115	58,852,319
BEKO	135,673	18,902,823	32,963	4,767,338	56	293,550	63,791	904,751			191	1,128,575	25,997,037
BOSCH	87,167	12,650,108	25,819	4,196,170	89	413,685	77,347	1,042,329			191	765,266	19,067,558
PROFILO	79,471	16,042,255	3,160	320,116	26	160,125							16,522,496
ARISTON	33,877	5,591,332	4,527	643,311	59	289,922	36,204	540,560					7,065,125
SIEMENS	27,191	3,640,153	9,668	1,423,413	120	503,072							5,566,638
ELECTROLUX			2,606	274,338	66	321,750	38,512	500,379					1,096,467
TELEFUNKEN	15,753	2,250,418	14,459	1,946,444	1	4,313	13,221	184,957					4,386,132
PANASONIC	8,847	589,148	4,185	599,478	55	373,802							1,562,428
CASPER	16,193	4,821,192	26,519	2,508,574	18	40,745	10,178	150,528					7,521,039
DELL			8,017	915,250	4	11,025							926,275
ACER			2,528	247,282									247,282
ARC PC			564	53,843	4	7,500							61,343
OTHER	131,532	16,780,210	56,360	7,972,296	388	1,601,162	63,777	908,805	70	6,456	156	519,314	27,788,243
TOTAL	**909,884**	**137,875,913**	**318,279**	**42,696,025**	**1,050**	**4,864,537**	**1,084,323**	**15,409,386**	**6,490**	**680,308**	**1,134**	**6,839,270**	**208,365,440**

KOÇ UNIVERSITY
GRADUATE SCHOOL OF BUSINESS
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Exhibit 17 . Order Flow Chart



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Exhibit 18. Manugistics Networks Transport

manugistics networks transport

manugistics

NetWORKS Transport

challenge:

Shipments occur every day within your global trading network.

Can every shipment, order, and line-item be located in real-time?

Are the entire organization's shipping requirements simultaneously considered and optimized?

Are carriers given continuous move opportunities?

Do you dynamically select crossdock and pool point locations based upon customer requirements and cost?

Can you manage and optimize multi-leg shipments to and from international locations?

Is the shipment execution process completely automated using EDI and Internet technologies?

Can you collaborate directly with each of your vendors?

Turn Transportation into a Strategic Advantage

Today, transportation is more critical to your success than ever before. As a result of the Internet, transportation is no longer viewed as a simple business function that must be managed, but rather as a strategic function that enables revenue generation and customer satisfaction. Couple the strategic nature of transportation with its relative percentage of an organization's total logistics cost — approximately 50% depending on industry — and it's no wonder transportation now garners a high-degree of attention in the logistics community.

NetWORKS Transport™ allows you to fully optimize your transportation operations by giving you the power to model and manage extremely detailed requirements and business rules. Backed by nearly two decades of transportation software experience and development, NetWORKS Transport offers a functional depth that is unmatched in the industry and includes capabilities that can make transportation a true profit center for your business.

Achieve Superior Transportation Management

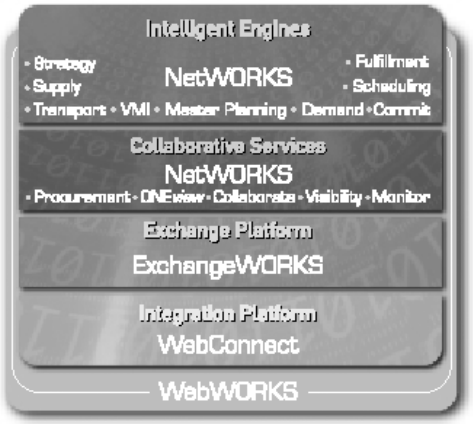
For example, NetWORKS Transport's multi-leg routing support allows an order to have many sub-shipments from origin to destination, considering and optimizing each individual leg while optimizing the entire multi-leg shipment. NetWORKS Transport can automatically recalibrate the in-process shipment and consider each throughpoint to automatically make the best service and cost decisions. NetWORKS Transport offers this level of depth and breadth of functionality to enable superior transportation management by making it possible for you to:

- cross-enterprise optimization
- optimize enterprise-wide
- automate the execution of the shipping process
- augment your solution options through integration to external exchanges
- track shipping results in real-time and historically
- create exception-based alerting
- receive, match, and pay freight bills
- facilitate dynamic or historical performance reporting

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manugistics networks transport

Transforming Supply Chains into Intelligent eBusiness Trading Networks



Provide increased revenue, lower operating costs, and increased customer satisfaction by:

- Simultaneously optimizing inbound, outbound, and inter-facility movements
- Tracking shipment progress to the customer
- Developing rules-based performance metrics and attaching real-time alerts to violations

Features & Benefits

- Supplier collaboration > **Improves item visibility and customer service**
- Global cross-enterprise optimization > **Decreases cost and increases customer service**
- Automated shipment execution > **Facilitates trading network visibility and exception-based shipment management via EDI and Internet communications**
- Freight payment > **Increases payment accuracy and reduces costs**

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Leveraged Intelligence

www.manugistics.com

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Vestel Durable Goods Marketing: Improving Distribution Management

Assignment:

In order to improve your understanding, first consider the exercise problem at the end of the case. Then prepare the following questions for discussion in the classroom.

- What are the key drivers of business for Vestel Durable Goods Marketing? What are important performance measures for the company and the supply chain? How have early SC distribution initiatives contributed to these drivers/measures?
- Evaluate the two transportation pricing schemes in terms of pros and cons for both parties involved. Identify incentive problems that could be encountered for each scheme.
- What would you do next if you were Levent Hatay? Identify the benefits and risks associated with the distribution-collaboration proposal, and the multiple-cargo-providers idea. How would you implement these initiatives?
- What are the differences between the relationship between Vestel and Horoz Logistics and regular 3PL relationships?
- Summarize the results obtained from the implementation of MTM, for Vestel. Quantify these as much as you can.

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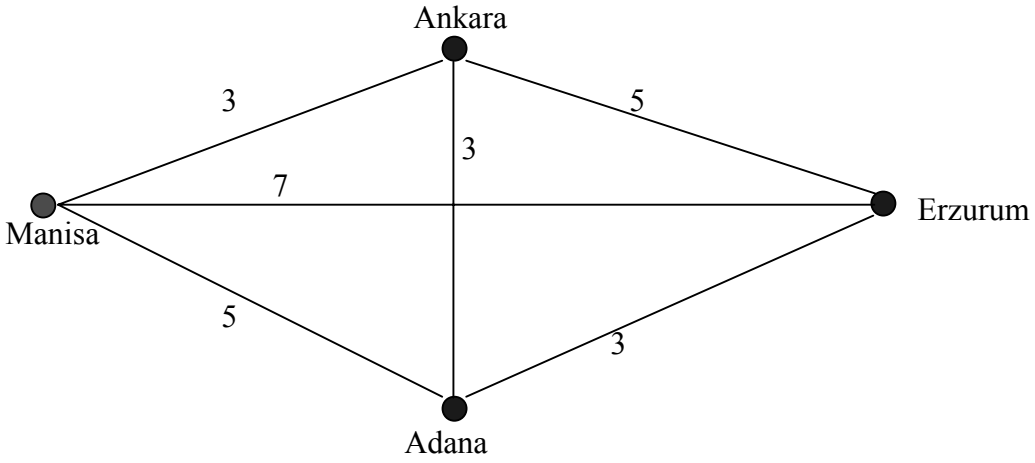
Vestel Durable Goods Marketing: Improving Distribution Management B

Exercise

Consider a simplified version of the distribution problem of Vestel. There are only two products: televisions (1 vol./unit) and refrigerators (3 vol./unit). There are three customers A, B, C located in Ankara, Adana, and Erzurum respectively. The demands from these customers are given below:

	A (Ankara)	B (Adana)	**C (Erzurum)**
Television	1	4	**2**
Refrigerator	4	1	**2**

The locations of the customers are given in the following network. Distances between the cities are given on the arcs in distance units.



There are two size trucks available: a large size truck that has a capacity of 18 vol. and costs $10 per unit distance and a small size truck that has a capacity of 10 vol. and costs $6 per unit distance. The return distance is not considered in determining total cost. These trucks are operated by a 3PL logistics company and available at all the cities. There is no limitation for the number of trucks to be used. Only the volumes of products are important to load the trucks.

1. Find a distribution plan that will yield a low transportation cost for Vestel. What is the average utilization of the trucks for your plan?

2. Assume that the trucks are returning back empty from the destinations due to the imbalance of the traffic of goods to be transported from the East to the West and from the West to the East. What is the average utilization of the trucks for the owners and operators of these trucks?

3. Another 3PL company offers a fixed price of $3.5 per television and $11 per refrigerator regardless of the destination. Would you accept this offer?

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Now consider the same problem for Bosch. Bosch is also delivering to their dealers in Ankara, Adana, and Erzurum. On the same day, their demands are given in the following table:

Bosch

	A (Ankara)	B (Adana)	**C (Erzurum)**
Television	2	1	**4**
Refrigerator	2	4	**1**

3- What happens if you coordinate the distribution of Vestel and Bosch? What will be the total saving?

4- How would you distribute the saving between Vestel and Bosch?

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